A Multi-Agent Approach to the Adaptation of Migration Topology in Island Model Evolutionary Algorithms

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***Abstract*—The Island Model (IM) is an important multi-population approach for improving the performance of Evolutionary Algorithms (EAs) when solving complex problems. One of the critical parameters for defining a suitable IM is the migration topology, which determines the migratory flows between sub-populations of the model. Despite the importance of this parameter, the majority of topologies tend to be naive and fail to take into account the underlying optimization process. To deal with the problem of adequately setting a migration topology, we propose an approach in which the Island Model is transformed into a Multi-Agent System (MAS) capable of learning and adapting the inter-island links based on the experience obtained during the evolutionary process. This approach is compared against two other traditional topologies applied to island versions of two different EAs, and to their usual implementations. The preliminary results strongly suggest an advantage of the IM versions over the original algorithms, and the competitiveness of the proposed approach.**

***Keywords*-Parallel Evolutionary Algorithms; Island Model; Migration Topologies; Reinforcement Learning; Multi-agent Evolutionary Algorithms**

I. Introduction

The Island Model (IM) [1] is a multi-population approach for Evolutionary Algorithms (EAs). The first studies about the IM were conducted by Grefenstette *et al.* [2], [3], and despite being initially proposed for Genetic Algorithms (GAs) it has been successfully applied to other evolutionary algorithms such as Differential Evolution (DE) [4] and Evolution Strategies (ES) [5].

In the IM implementation of Evolutionary Algorithms, the population is divided into subpopulations, called *islands*, which independently execute the EA and occasionally exchange individuals, in a process called *migration*. The migration topology describes the links between the islands and plays an important role in the performance of the IM[6].

Despite its importance, it is usual for the migration topologies employed in the literature to be naive and based mainly on computer networks and architectures. These static topologies do not take into account the dynamics of the underlying exploration, and ignore the knowledge obtained during the search process.

In this paper, we propose a new approach to automatically set the migration topology, with the goal of increasing the performance and flexibility of the IM. The proposed approach transforms the IM into a multi-agent system, generating and adapting the migration topologies based on a reinforcement learning method called Q-learning [7].

Comparative tests between the reinforcement learning approach and two migration topologies widely used in the literature, namely the Ring [1] and Random [8] topologies, were conducted. Island Model versions of the optimization methods GL-25 (Global and Local Real-coded Genetic Algorithms) [9] and SaMDE (Self-adaptive Mutation in the Differential Evolution) [10] were implemented for these experiments. The preliminary results suggest that the proposed approach is promising, showing competitive performance when compared with traditional migration topologies.

The remainder of this paper is organized as follows: Section II presents the Island Model and briefly describes and analyzes the ring and random approaches. Section III describes the Q-learning method; Section IV presents the proposed approach; Section V presents and discusses the comparative results obtained in a well known set of global optimization benchmark problems. Section VI shows some topologies obtained by the proposed approach, and finally Section VII concludes the paper and presents some final considerations.

II. Background

A. Island Model

The Island Model (IM) is a popular and efficient way to implement Evolutionary Algorithms on both serial and parallel machines [1], [4]. The main idea behind the IM is to divide the population of solutions into subpopulations relatively isolated, called islands. The relative isolation of the islands fosters a competition between them.

Although competition is a positive point, collaboration is fundamental to the goal of finding the best solution in

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an optimization problem. This collaboration is promoted by a periodical migration process, in which individuals with different characteristics are inserted in other subpopulations. Migration allows higher fitness subpopulations to aid less developed ones by introducing new information through immigrants.

The main benefit of this model is the improvement of the search in the solution space, balancing exploration and exploitation [6]. Despite of being suitable for parallel implementation models, the IM is also beneficial to the evolutionary search itself, since some enhancement in solution quality and convergence time can be reached even in sequential implementations [4].

For a good behavior of the IM, the developer must set some parameters which are directly related to the performance of the model. These parameters are:

1) **Number of islands:** defines the number of islands or subpopulations in the model.
2) **Migration topology:** defines the communication structure of the islands, i.e., the links between subpopulations.
3) **Migration frequency:** defines how often the migrations occur.
4) **Migration rate:** defines how many individuals or solutions migrate from a subpopulation to another.
5) **Synchronization type:** defines the type of synchronization for the migration process (synchronous or asynchronous).
6) **Migration policy:** defines the individuals or solutions which will be removed and replaced.

B. Topologies

Migration topology is the name given to the set of edges that determine the neighborhood structure of the island model. The neighborhood is represented by connections of edges, interconnecting subpopulations that are allowed to interact directly. Through this communication network the islands can exchange individuals, and this exchange can be either unidirectional or bidirectional.

A migration topology usually found in the literature is the Ring Topology[1], in which the subpopulations are interconnected in the form of a closed circle. Its rigid format does not incorporate any knowledge about the search process behavior. Although simple, the Ring Topology can be a weak strategy in a IM with a large number of subpopulations, because it may take many generations to transmit good individuals to all islands.

On the other hand, the Random Topology[8] is totally dynamic. In this schema a new topology is defined randomly each time the migration has to occur. Despite being a more flexible solution for the migration topology, once more, it does not consider any knowledge about the search process.

III. Q-LEARNING

Q-learning [7] is a stochastic method of reinforcement learning, which models learning through trial-and-error interactions with a dynamic environment [11]. In this method, the agent/learner takes an action by itself and receives a corresponding reward or punishment. The process of learning emerges from the maximization of the rewards by the agent. The learning of the best actions in each state is possible after applying a sequence of actions and observing the payoffs of each action performed. The method learns the optimal policy using (1):

$$Q(s_t, a_t) \leftarrow \lambda(r + \gamma \max_{\alpha'} Q(s_{t+1}, a_{t+1})) + (1 - \lambda)Q(s_t, a_t) \tag{1}$$

where, s_t is the current state; a_t is the action performed in s_t; r is the reward received after taking the action a_t; s_{t+1} is the new state; γ is a discount factor; and λ is the learning rate.

Initially, the method randomly chooses the state-action pairs. Afterwards, the algorithm tries to determine the best actions from a defined state, estimating the rewards through the past experiences and estimates of future rewards. The pseudo-code of the Q-learning method is presented in Algorithm 1.

Algorithm 1: Q-learning Algorithm

```
INPUT
λ ← learning rate , γ ← discount factor, ϵ ←
exploration rate ;
BEGIN
∀s ∀a Q(s,a) ← 0;
s_t ← current state;
if rand() < ϵ then
|  a_t ← random action;
else
|  a_t ← arg max_α Q(s_t,a_t);
end
Take action a_t;
Receive reward r;
s_{t+1} ← current state;
Q(s_t,a_t) ← λ(r + γ max_α' Q(s_{t+1},a_{t+1})) + (1 -
λ)Q(s_t,a_t);
λ ← 0.99*λ;
ϵ ← 0.98*ϵ;
goto 6;
END
```

The learning rate, discount factor and exploration rate are some of the Q-learning parameters. The learning rate determines how much value is given to new information over old information. The discount factor determines the relevance of future rewards. Finally, the exploration rate defines if the agent will take the best known action at the moment or if it will explore the actions space. Notice, that these parameters can influence the results.

IV. Adaptation of the Migration Topology

As mentioned in Section II, the migration topology plays an important role in the IM performance. With this in mind, an approach that uses Q-learning to dynamically adapt the topology is proposed . The idea is to consider the Island Model as a learning system, which will be able to learn the inter-island links from the experience obtained during the search process. As the quality of the solutions in each island changes, this system will also be able to adjust these links in a way that enhances the evolutionary process.

In this approach each subpopulation or island is associated with a learning agent. Each agent attempts to learn which subpopulation contributes the most to its evolution, establishing links with these subpopulations. Thus, the topology will emerge from the "beliefs" of each island, about which is the best neighbor it can have. It is important to notice that the learning process is independent for each island.

In the models built by Q-learning the states are the islands from which a defined agent can receive individuals on the migration process. Each state is associated with only one island of the model. The actions represent the exchange or continuation of the subpopulation chosen to send solutions.

Figure 1 depicts the operation of an agent. In this diagram, the islands are the subpopulations and the blue dashed arrows are the reachable states of Agent 4. The blue solid arrow represents the current state of Agent 4 and means that the Island 4 is receiving individuals from Island 2. The red arrows represent the possible actions for Agent 4 at the current state. If Agent 4 takes Action 1, Island 4 will keep receiving individuals from Island 2; if, on the other hand, Agent 4 takes Action 3, Island 4 will change its migration flow by stopping the inflow of individuals from Island 2 and starting receiving individuals from Island 1. The operation of other agents works the same way.

Figure 2 illustrates what would be the payoff matrix of the situation above. The states represent the origin of the individuals and the actions tell if there will be a change on the migratory flow, that is, if the island in question will receive individuals from somewhere else. Actions with higher values are the ones that the agents "believe" to be the best ones at the moment, given the current state.

The number of states is defined by Num_States = ($Num\_Islands$ - 1), where $Num\_Islands$ represents the number of subpopulations. The number of possible actions are $Num\_Actions$ = $(Num\_States)^2$.

After defining the migration topology, the migration process is executed and for each island the payoff is calculated (2):

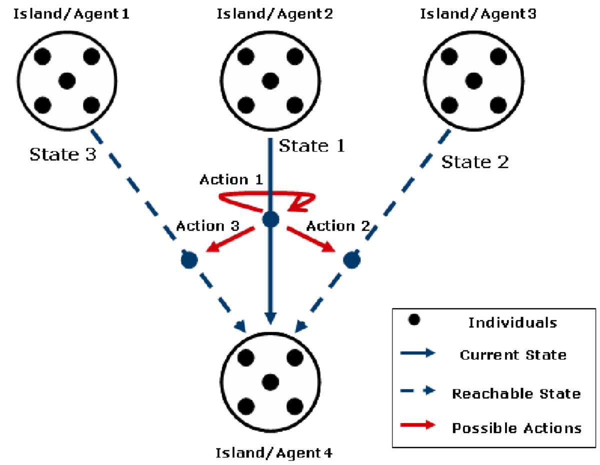


Figure 1. Agent operation.

		Actions		
	States\Actions	Receive from Island 1	Receive from Island 2	Receive from Island 3
States	Receiving from Island 1	0.1456	0.1609	0.0912
	Receiving from Island 2	0.1544	0.1393	0.2023
	Receiving from Island 3	0.0635	0.1196	0.5112

Figure 2. Payoff matrix of Agent 4 for a island model with 4 islands.

$$p = \frac{(CurrentFitness - ReceiveFitness)}{\max(CurrentFitness, ReceiveFitness)} \quad (2)$$

where, $CurrentFitness$ is the fitness or cost function value of the best solution in the current subpopulation and $ReceiveFitness$ is the cost function of the solution received from the island determined by the Q-learning agent. Notice that (2) implies in a minimization optimization problem.

The pseudo-code of the proposed approach is presented in Algorithm 2.

V. Comparative Tests

In this section the proposed approach is compared with the Ring Topology and the Random Topology, described in Section II-B. For this comparison the IM versions of the optimization methods GL-25[1] (Global and Local Real-coded Genetic Algorithm) [9] and SaMDE (Self-adaptive Mutation in the Differential Evolution) [10], were implemented, using each of the approaches to configure their topologies. The original versions of the algorithms were also tested. Therefore, for each optimization algorithm we have the following configurations:

[1]In this paper the single population version of GL-25 was taken from http://dces.essex.ac.uk/staff/qzhang/. Minor modifications were made in order to adapt it to the island version.

Algorithm 2: Q-learning Topology

INPUT
$\lambda \leftarrow$ learning rate $\gamma \leftarrow$ discount factor $\epsilon \leftarrow$ exploration rate;
$GEN \leftarrow$ migration frequency;
BEGIN
for *each subpopulation* **do**
    Initialize the initial state of Q-learning agent randomly;
    Initialize subpopulation;
end
while $\neg$ $stop\_condition$ **do**
    for *each subpopulation* **do**
        Apply the optimization algorithm (GEN);
        Select the action of Q-learning agent;
        Apply the migration process;
        Estimate the reward;
        Update the Q-learning agent;
    end
end
END

- Classic: single population version of the optimization method;
- Q-learning: IM version using Q-learning to configure the migration topology;
- Rand: IM version with random topology;
- Ring: IM version with ring topology.

On the IM versions the best individual of each subpopulation migrates and substitutes a random individual of the receiving subpopulation. The migrations occur every 20 generations and the number of individuals per island is 100. The classic versions have the number of individuals equal to the number of individuals per island of the IM versions times the number of islands. The parameters used for the Q-learning were $\gamma = 1.0$, $\lambda = 0.5$ and $\epsilon = 1.0$. Both the IM parameters and the Q-learning parameters were defined based on a few preliminary runs.

In these experiments, the 6 test functions of the Benchmark Functions for Large Scale Global Optimization [12] that have known optima were taken. These functions are described in Appendix A. Functions f_1 and f_2 are unimodal, and f_3 to f_6 are multimodal. All functions were defined with 100 variables.

Each configuration of the algorithms was executed on 15 independent runs for each function, and the search was terminated when either one of the following criteria was satisfied:

1) Maximum number of generations has been reached;
2) $\Delta f \leq 0.00001$ is found[2].

The results are separated by optimization algorithm and are reported below.

A. GL-25

In this section the results concerning the GL-25 are presented. Figures 3, 4 and 5 depict the empirical cumulative distribution of *runtimes*[3] from the different approaches to the topology configurations for the IM with 4, 8 and 16 islands, respectively. A Kruskal-Wallis test detected significant differences among the final convergence proportions ($p < 0.05$), and post-hoc comparisons (Wilcoxon Rank-Sum tests) were able to detect differences between the island approaches and the classic version, but no significant difference among the island versions. One must notice, however, that for the cases with 8 and 16 islands the Q-learning approach was able to obtain the highest percentage of convergence (albeit not significantly superior to the other island-based configurations) with less function evaluations.

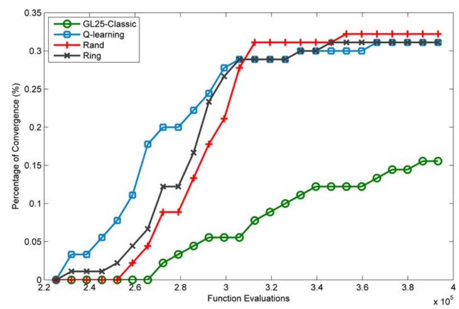


Figure 3. GL-25 - 4 islands.

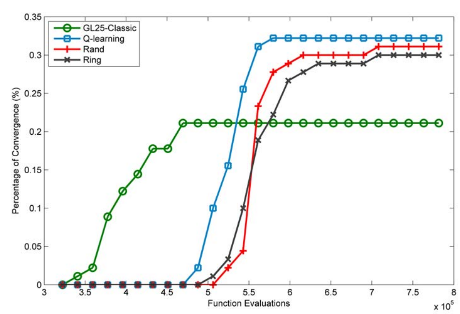


Figure 4. GL-25 - 8 islands.

[2] Δf is target precision value. It is defined by $\Delta f = f_{target} - f_{optimum}$

[3] Runtime is defined here as the number of function evaluations until the optimal solution (with a certain precision) was reached.

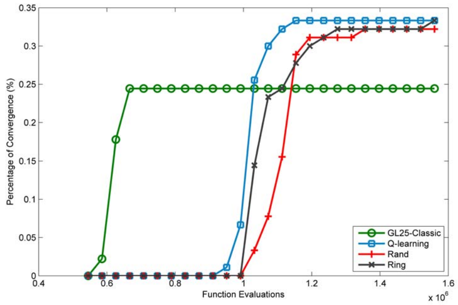


Figure 5. GL-25 - 16 islands.

B. SaMDE

In this section the results concerning the SaMDE algorithm [10] are presented. Figures 6, 7 and 8 depict the empirical cumulative distribution of runtimes from the different topology configuration approaches for the IM with 4, 8 and 16 islands, respectively. Similarly to the GL-25 results, a Kruskal-Wallis test found significant differences at the 95% confidence level, with the results obtained by the IMs significantly better than those obtained by the classic version.

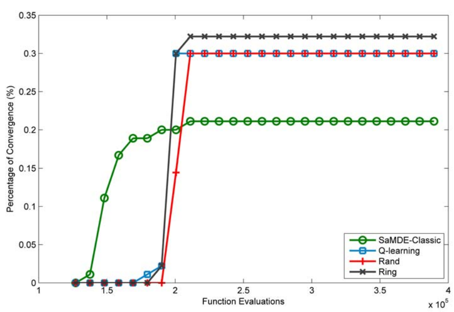


Figure 6. SaMDE - 4 islands

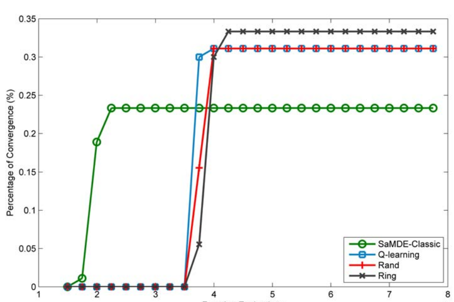


Figure 7. SaMDE - 8 islands

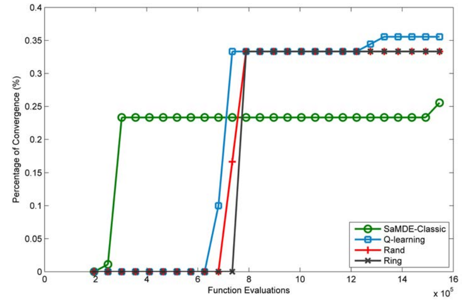
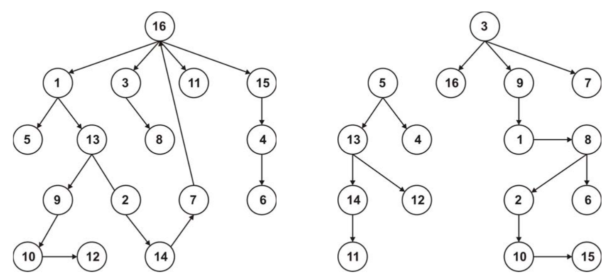


Figure 8. SaMDE - 16 islands

VI. Resulting Topologies

This section illustrates some of the topologies obtained by the proposed Q-learning approach. It is important to notice that, in the beginning of the optimization process there is a learning period during which the topologies constantly change. The ones illustrated here correspond to the final topologies used by the algorithms. Due to space constraints, only the results of unimodal function f_1 and multimodal function f_5 for the IM with 16 islands are shown.

Figures 9 and 10 depict the topologies obtained by GL-25 and SaMDE respectively. It can be seen that the structures obtained for the unimodal function are more compact when compared with the structures obtained for the multimodal one. This is an interesting feature because the dissemination of good solutions in a more compact structure is faster, and this can increase the convergence speed of the algorithm. On the other hand, for multimodal functions a compact structure is more likely to get stuck in local minima, with less compact structures decreasing the chances of premature convergence, since good solutions (which can be local minima) take more time to spread to the other subpopulations.

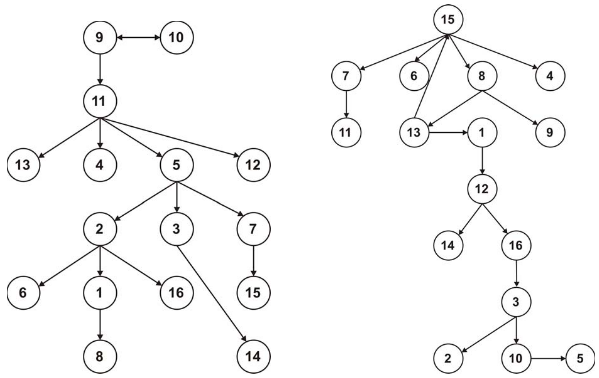


(a) Unimodal Function f_1 (b) Multimodal Function f_5

Figure 9. GL-25 (16 Islands)

VII. Conclusions

In this paper, a new approach to generate and adapt the migration topology in EAs using the Island Model was presented and discussed. In the proposed approach the Island Model is transformed into a multi-agent system which learns



(a) Unimodal Function f_1 (b) Multimodal Function f_5

Figure 10. SaMDE (16 Islands)

and adapts the inter-island links based on the experience obtained during the evolutionary process.

The proposed approach presented promising features, such as the apparent adaptation to the function being optimized, with compact structures for unimodal functions and less compact structures for multimodal ones. These preliminary results point to interesting possibilities in the use of IMs. Moreover, the results also suggest that the proposed approach is competitive when compared with the traditional topologies, particularly in the convergence speed. However, more extensive testing is necessary in order to obtain a more quantitative knowledge of the performance of proposed approach. In particular, more sophisticated experimental designs can be employed to isolate the effects of function modality, dimension, and underlying algorithm in the performance of the IM approaches.

Besides refining the experimental analysis, future works include the incorporation of a diversity mechanism in the learning process, in order for migration to encourage the diversification of the population; and the implementation of the proposed model in parallel computer environments.

APPENDIX

This appendix describes the test functions used in the experimental analysis.

$$f_1(x) = \sum_{i=1}^{D} x_i^2, \qquad x_i \in [-100, 100] \tag{3}$$

$$f_2(x) = \max\{|x_i|, 1 \leq i \leq D\}, \qquad x_i \in [-100, 100] \tag{4}$$

$$f_3(x) = \sum_{i=1}^{D-1} (100(x_i^2 - x_{i-1})^2 + (x_i - 1)^2), \qquad x_i \in [-100, 100] \tag{5}$$

$$f_4(x) = \sum_{i=1}^{D} (x_i^2 - 10\cos(2\pi x_i) + 10), \qquad x_i \in [-5, 5] \tag{6}$$

$$f_5(x) = \sum_{i=1}^{D} \frac{x_i^2}{4000} - \prod_{i=1}^{D} \cos(\frac{x_i}{\sqrt{i}}) + 1, \qquad x_i \in [-600, 600] \tag{7}$$

$$f_6(x) = -20\exp(-0.2\sqrt{\frac{1}{D}\sum_{i=1}^{D} x_i^2}) - \exp(\frac{1}{D}\sum_{i=1}^{D}\cos(2\pi x_i)) + 20 + e, \qquad x_i \in [-600, 600] \tag{8}$$

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